

Lab 1 :

The screenshot shows a PostgreSQL query editor window titled 'dvdrental/postgres@PostgreSQL 18'. The 'Query' tab is active, displaying a SQL script with 14 lines. The script creates a role, two users, grants privileges, and sets up row-level security. The 'Messages' tab is selected, showing the execution of the first line, 'CREATE ROLE', which completed successfully in 71 msec. A green status bar at the bottom indicates the query was successful.

```
1 CREATE ROLE cashier_role;
2 CREATE USER vianh WITH PASSWORD 'via123';
3 CREATE USER thu WITH PASSWORD 'thu123';
4 GRANT cashier_role TO vianh;
5 GRANT cashier_role TO thu;
6
7 GRANT SELECT ON film, staff TO cashier_role;
8 GRANT SELECT, INSERT ON payment TO cashier_role;
9
10 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
11 CREATE POLICY
```

CREATE ROLE

Query returned successfully in 71 msec.

✓ Query returned successfully in 71 msec. ✕

Total rows: Query complete 00:00:00.071 CRLF Ln 2, Col 1

This screenshot is identical to the one above, showing the same SQL script and successful execution of the first line. The status bar at the bottom indicates the query was successful in 81 msec.

```
1 CREATE ROLE cashier_role;
2 CREATE USER vianh WITH PASSWORD 'via123';
3 CREATE USER thu WITH PASSWORD 'thu123';
4 GRANT cashier_role TO vianh;
5 GRANT cashier_role TO thu;
6
7 GRANT SELECT ON film, staff TO cashier_role;
8 GRANT SELECT, INSERT ON payment TO cashier_role;
9
10 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
11 CREATE POLICY
```

CREATE ROLE

Query returned successfully in 81 msec.

✓ Query returned successfully in 81 msec. ✕

Total rows: Query complete 00:00:00.081 CRLF Ln 3, Col 40

The screenshot shows a PostgreSQL query editor window titled 'dvdrental/postgres@PostgreSQL 18'. The query editor contains the following SQL statements:

```
1 CREATE ROLE cashier_role;
2 CREATE USER vianh WITH PASSWORD 'via123';
3 CREATE USER thu WITH PASSWORD 'thu123';
4 GRANT cashier_role TO vianh;
5 GRANT cashier_role TO thu;
6
7 GRANT SELECT ON film, staff TO cashier_role;
8 GRANT SELECT, INSERT ON payment TO cashier_role;
9
10 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
11 CREATE POLICY
```

The 'Data Output' tab is selected, showing the following message:

```
GRANT ROLE
Query returned successfully in 73 msec.
```

A green status bar at the bottom right indicates: 'Query returned successfully in 73 msec. X'. The bottom status bar shows 'Total rows: Query complete 00:00:00.073', 'CRLF', and 'Ln 5, Col 27'.

The screenshot shows a PostgreSQL query editor window titled 'dvdrental/postgres@PostgreSQL 18'. The query editor contains the following SQL statements:

```
1 CREATE ROLE cashier_role;
2 CREATE USER vianh WITH PASSWORD 'via123';
3 CREATE USER thu WITH PASSWORD 'thu123';
4 GRANT cashier_role TO vianh;
5 GRANT cashier_role TO thu;
6
7 GRANT SELECT ON film, staff TO cashier_role;
8 GRANT SELECT, INSERT ON payment TO cashier_role;
9
10
11
```

The 'Data Output' tab is selected, showing the following message:

```
GRANT
Query returned successfully in 74 msec.
```

Lab 2:

```
dvdrental/postgres@PostgreSQL 18
Query Query History Scratch Pad x
1 CREATE ROLE cashier_role;
2 CREATE USER vianh WITH PASSWORD 'via123';
3 CREATE USER thu WITH PASSWORD 'thu123';
4 GRANT cashier_role TO vianh;
5 GRANT cashier_role TO thu;
6 GRANT SELECT ON film, staff TO cashier_role;
7 GRANT SELECT, INSERT ON payment TO cashier_role;
8
9 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
10
11
12
13
Data Output Messages Notifications
ALTER TABLE
Query returned successfully in 70 msec.
Total rows: Query complete 00:00:00.070 CRLF Ln 9, Col 47
✓ Query returned successfully in 70 msec. ✕
```

```
dvdrental/postgres@PostgreSQL 18
Query Query History Scratch Pad x
3 CREATE USER thu WITH PASSWORD 'thu123';
4 GRANT cashier_role TO vianh;
5 GRANT cashier_role TO thu;
6 GRANT SELECT ON film, staff TO cashier_role;
7 GRANT SELECT, INSERT ON payment TO cashier_role;
8
9 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
10 CREATE POLICY isolate_cashier_payments ON payment
11 FOR SELECT
12 TO cashier_role
13 USING (
14     staff_id = (SELECT staff_id FROM staff WHERE lower (username) = curi
15 );
16
17
18
Data Output Messages Notifications
CREATE POLICY
Query returned successfully in 66 msec.
Total rows: Query complete 00:00:00.066 CRLF Ln 15, Col 3
✓ Query returned successfully in 66 msec. ✕
```

Dashboard x Properties x SQL x Statistics x Dependencies x Processes x dvdrental/postgres@PostgreSQL 18 x

dvdrental/postgres@PostgreSQL 18

Query Query History Scratch Pad x

```
4 GRANT cashier_role TO vianh;
5 GRANT cashier_role TO thu;
6 GRANT SELECT ON film, staff TO cashier_role;
7 GRANT SELECT, INSERT ON payment TO cashier_role;
8
9 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
10 CREATE POLICY isolate_cashier_payments ON payment
11 FOR SELECT
12 TO cashier_role
13 USING (
14     staff_id = (SELECT staff_id FROM staff WHERE lower (username) = curi
15 );
16 SET ROLE thu;
17 SELECT COUNT(*) FROM payment ;
18
19
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	count	bigint
1		0

Total rows: 1 Query complete 00:00:00.083 CRLF Ln 17, Col 20

dvdrental/postgres@PostgreSQL 18

Query Query History Scratch Pad x

```
4 GRANT cashier_role TO vianh;
5 GRANT cashier_role TO thu;
6 GRANT SELECT ON film, staff TO cashier_role;
7 GRANT SELECT, INSERT ON payment TO cashier_role;
8
9 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
10 CREATE POLICY isolate_cashier_payments ON payment
11 FOR SELECT
12 TO cashier_role
13 USING (
14     staff_id = (SELECT staff_id FROM staff WHERE lower (username) = curi
15 );
16 SET ROLE thu;
17 SELECT COUNT(*) FROM payment ;
18 SET ROLE vianh;
19 SELECT COUNT(*) FROM payment ;
20
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	count	bigint
1		0

✓ Successfully run. Total query runtime: 183 msec. 1 rows affected. ✕

Total rows: 1 Query complete 00:00:00.183 CRLF Ln 19, Col 31